

How to write your introduction

General concepts for writing

- Look up an article that is related to what your project – may help to follow that format or help to find relevant references
- Write the manuscript in a simple and accessible style – write for other scientists, avoid unnecessary simplification – BUT many readers are outside the immediate discipline, so clarity of expression is needed
- Avoid jargon and acronyms where possible - when essential, they should be defined at first use

Logistics of writing an Introduction

Four sections:

1. Context/importance
2. Background/overview of field
3. Current understanding or current tool, AND gaps in knowledge or inadequacies of tools
4. Summary of your study

Introduction: paragraph 1: Context / Importance

- The first paragraph should be fairly short and state the big picture problem that your project addresses.
 - If your project addresses a disease, describe the impacts of the disease
 - If your project generates a biological tool, describe the general biology knowledge and/or applications that could be impacted by that tool
 - If your project generates a bioinformatics tool, describe the general biology knowledge that could be impacted by that tool and/or importance of the data that is being analyzed
- *Importantly* end this first paragraph with **a key sentence** to state the general gap in knowledge or general lack of solutions (this will hint at what your project seeks to address).

Structure of the human astrovirus capsid spike in complex with the neonatal Fc receptor

Human astroviruses (HAstVs) are an important cause of viral diarrhea in children worldwide^{1–3}. Between 2–9% of non-bacterial diarrhea in children can be attributed to HAstVs^{4,5}. Seropositivity rates for HAstVs are high, with 90% of the population having antibodies towards at least one HAstV by the age of 9^{6–30}. In addition to the classical HAstVs (HAstV1–8), the divergent HAstVs (MLB and VA/HMO clades) are also found in human stool samples worldwide^{11–15}. Despite their global prevalence, HAstVs remain relatively poorly characterized, impeding the development of HAstV vaccines and therapeutics.

HAstVs are non-enveloped icosahedral viruses with positive-sense single-stranded RNA genomes. Their genomes range from 6–8 kilobases and contain three open reading frames, ORF1a, ORF1b, and ORF2. ORF1a and ORF1b encode nonstructural proteins, and ORF2 encodes the capsid protein. The classical HAstV capsid protein precursor has a highly basic amino terminus that binds to the viral genome, a core domain that encapsulates and protects the viral genome, a spike domain that forms dimeric globular spikes on the surface of the

METHODS

Inferring multi-locus selection in admixed populations

Nicolas M. Ayala^{1,2}, Maximilian Genetti^{1,2}, Russell Corbett-Detig^{1,2*}

1 Genomics Institute, University of California, Santa Cruz; Santa Cruz, California, United States of America,

2 Department of Biomolecular Engineering, University of California, Santa Cruz; Santa Cruz, California, United States of America

Introduction

Admixture is one of the primary sources of selected alleles in natural populations [1–3]. For example, in *Helianthus* sunflowers, introgressed alleles enhanced herbivore resistance at a number of loci [4]. In the fish *Fundulus grandis*, recently introgressed alleles allow resistance to extreme pollution and environmental change [5]. In humans, introgression from archaic hominids is thought to have facilitated adaptation to a range of novel environments [6,7]. Similarly, admixture can contribute alleles that are not adapted to local environments (e.g., as in [8]) or it may contribute haplotypes that are deleterious due to accumulation of weakly deleterious mutations during long term isolation in small populations [9], and are therefore purged by natural selection. Finally, in some cases admixed populations may contain mutations contributed by separate parental populations that have negative interactions thereby resulting in strong selection within the admixed populations [8,10]. Although the importance of selection on admixed ancestry, or adaptive introgression, is increasingly appreciated, generalized methods to accurately detect and quantify the impacts of natural selection from genome sequence data in admixed populations are in their infancy.

Introduction section: section 2: Background / Overview of field

- The next section should give an overview of the field and describe the background necessary to fully understand your project.
- Some projects will require more than one paragraph for this section (i.e. more than one concept to introduce).
- This section should be comprehensive - but minimize adding information that is unnecessary or could be saved for the discussion section.
- If referring to your Figure 1, cite that figure at the end of the sentence (Fig. 1).

A quick note on referencing

- Be sure to add references at the end of any sentence that states a past research finding. Reference primary research articles (rarely review articles)
- If referencing a review article, acknowledge it – i.e. “Astroviruses form a large virus family that infect mammals and birds (reviewed in [4]).”
- *I will look up your references and see if they are relevant and real (beware of AI!)*
- Utilize a reference manager (Endnote, Zotero).
 - <https://guides.library.ucsc.edu/c.php?g=240807&p=1601208>
 - I don't require a specific style, but I prefer styles that add minimal characters/space to the text (i.e. ⁴ or [4] or not (Johnson et al. 2024))
 - References at the end should be easily searchable for me to look up (i.e. not: 1. Johnson et al. *Science*. 2024).

Introduction: section 3:

Current understanding or current tools, AND gaps in knowledge or inadequacies of tools

- The 3rd section should describe the finer details of your project's field – what is known, what tools are currently available
- Be sure to continue referencing primary research articles
- *Importantly* end this paragraph with **one or more sentences** to state the specific gaps in knowledge or lack of a solution that your project specifically addresses.

Structure of the human astrovirus capsid spike in complex with the neonatal Fc receptor

copies of the HAsV capsid to form the immature HAsV particle, presenting 90 dimeric spikes¹⁹. Extracellular proteolysis results in the removal of spikes and cleavage of the capsid surface²⁰, forming the mature HAsV particle, presenting only 30 dimeric spikes. While the reason for this loss of 60 spikes remains unclear, it may promote receptor binding and/or genome uncoating. Previous studies suggest that the HAsV capsid spike is involved in receptor engagement, and a number of conserved sites were proposed as potential receptor binding sites^{21–23}. In addition, neutralizing monoclonal antibodies (mAbs) targeting the HAsV spike can block virus attachment to human cells^{10,23–25}, further highlighting its potential involvement in receptor binding. However, without the identification of a receptor for HAsVs, these ideas have remained unconfirmed.

Recently, the neonatal Fc receptor (FcRn) was identified in two

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2 Department of Biomolecular Engineering, University of California, Santa Cruz; Santa Cruz, California, United States of America

The previous approaches are suitable for finding evidence of selection at a single site in an otherwise neutrally-evolving genome, but in general they do not account for cases where multiple selected sites are genetically linked and may affect the trajectories of each other due to linkage [19,20,43]. Existing methods are expected to incorrectly estimate the selection coefficients of individual sites when they are impacted by linkage with other selected alleles. This makes estimating the selection coefficient of each variant more difficult within admixed populations [43]. These methods also cannot distinguish between single and multiple site selection models, which may lead to an overestimation of the number of selected sites present on a chromosome within an admixed population in some circumstances. We therefore do not have the tools to investigate the impacts of multi-locus selection in admixed populations, but we expect that this phenomenon is widespread for the reasons we described above.

Introduction: section 4:

Summary of your study

- The 4th paragraph should describe what you discover or accomplish in your study
- Write in the active voice ("Here, we report...") as readers find concepts and results to be conveyed more clearly if written directly
- End this paragraph with **a sentence** to state how your project helps to address the general gap in knowledge or general lack of solutions **that you wrote in paragraph 1**.

Structure of the human astrovirus capsid spike in complex with the neonatal Fc receptor

End of paragraph 1:

found in human stool samples worldwide^{11–15}. Despite their global prevalence, HAsTVs remain relatively poorly characterized, impeding the development of HAsTV vaccines and therapeutics.

Final paragraph:

Here, we report the crystal structure of FcRn in complex with the HAsTV1 capsid spike domain. This structure defines the molecular interactions governing HAsTV binding to FcRn, revealing a previously unrecognized site of vulnerability on the surface of the HAsTV virion. Binding affinity studies provide functional insights into the breadth and pH dependence for FcRn binding, and competition binding studies provide insights into the mechanism of antibody neutralization. Altogether, these studies increase our understanding of HAsTV infection and provide a foundation for the development of a HAsTV vaccine and antiviral therapies.

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² Department of Biomolecular Engineering, University of California, Santa Cruz; Santa Cruz, California, United States of America

End of paragraph 1:

strong selection within the admixed populations [8,10]. Although the importance of selection on admixed ancestry, or adaptive introgression, is increasingly appreciated, generalized methods to accurately detect and quantify the impacts of natural selection from genome sequence data in admixed populations are in their infancy.

Final paragraph :

We introduce an approach for modeling the effects of linkage between multiple selected sites within admixed populations of eukaryotic sexually recombining species. We validated our method under a variety of simulated scenarios, where the introgressing population introduced multiple alleles under selection. This approach can accurately identify the number of linked selected sites, as well as determine their location and estimate their selection coefficients by considering the impacts of the linked selected alleles. We applied our method to an admixed population of *D. melanogaster*, and we show that our previous method may have overestimated both the number of selected sites and their selection coefficients due to the effects of linkage [43]. We also applied our method to an admixed population of *P. italica*, and similarly found that fitting single site models may overestimate the selection coefficients of selected positions when compared to multi-locus models. Our results suggest that this is an important contributor to evolutionary outcomes in admixed populations, and our work provides a powerful generalized tool to quantitatively investigate these effects.